PhD Qualification Exam in Applied Mathematics

Mock Examination No. 3

Total: 100 points

Problems 1–5 are required. Solve exactly one of Problems 6 and 7. If both are attempted, only the higher score will be counted.

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1. (15 points) For the harmonic oscillator $q' = v$, $v' = -\omega^2 q$ with $\omega > 0$, consider the Störmer–Verlet method

$$v_{n+1/2} = v_n - \frac{h}{2}\omega^2 q_n, \qquad q_{n+1} = q_n + h v_{n+1/2}, \qquad v_{n+1} = v_{n+1/2} - \frac{h}{2}\omega^2 q_{n+1}.$$

Derive the amplification matrix, prove second-order accuracy, and find the exact linear stability condition. Show that the map has determinant one and determine its numerical frequency in the stable regime.

2. (15 points) Let $A = A^T$ have a simple extreme eigenpair (λ_1, q_1) separated from the other eigenvalues (that is, λ_1 is either the largest or the smallest eigenvalue). Rayleigh quotient iteration is

$$\mu_k = x_k^T A x_k, \qquad (A - \mu_k I) y_k = x_k, \qquad x_{k+1} = y_k / \|y_k\|_2.$$

Assume $\|x_k\|_2 = 1$ and x_k is sufficiently close, but not equal, to q_1. Prove local cubic convergence of the angle between x_k and q_1. State precisely where symmetry and the spectral gap are used.

3. (20 points) Consider the mean-zero periodic problem

$$-u'' = f \quad \text{on } [0, 2\pi], \qquad u(0) = u(2\pi), \quad u'(0) = u'(2\pi), \qquad \int_0^{2\pi} u \, dx = 0.$$

Let $V_N = \operatorname{span}\{e^{ikx} : 0 < |k| \le N\}$.

(a) State and prove the solvability condition for the continuous problem and give its Fourier solution.

(b) Formulate the Fourier Galerkin method in V_N and derive its coefficients and computational complexity when FFT data are available.

(c) If $f \in H^s$ has zero mean, prove spectral projection estimates for $\|u - u_N\|_{L^2}$ and $\|(u - u_N)'\|_{L^2}$ with explicit powers of N.

4. (15 points) For $u_t + a u_x = 0$, $a > 0$, discretize $-a u_x$ by the backward upwind difference and advance the resulting system by the two-stage SSP Runge–Kutta method

$$U^{(1)} = U^n + \tau L_h U^n, \qquad U^{n+1} = \tfrac{1}{2} U^n + \tfrac{1}{2}\big(U^{(1)} + \tau L_h U^{(1)}\big).$$

Determine the consistency order. Prove an ℓ^∞ maximum principle and ℓ^2 stability under a sharp CFL condition, and give the Fourier amplification factor.

5. (15 points) For $u_t = v$, $v_t = u_{xx}$ with homogeneous Dirichlet boundary conditions, consider the implicit midpoint discretization

$$\frac{U^{n+1} - U^n}{\tau} = \frac{V^{n+1} + V^n}{2}, \qquad \frac{V^{n+1} - V^n}{\tau} = \delta_{xx} \frac{U^{n+1} + U^n}{2}.$$

Determine the truncation order and prove conservation of an appropriate discrete energy. Deduce unconditional stability.

6. (20 points; choose one of Problems 6–7) For $0 < \varepsilon \ll 1$, consider

$$\varepsilon y'' + (1+x)y' = 1, \quad 0 < x < 1, \qquad y(0) = y(1) = 0.$$

Locate the boundary layer and determine its thickness by dominant balance. Construct a uniformly valid composite expansion through formal order ε, and state its expected accuracy with a residual and boundary-condition check.

7. (20 points; choose one of Problems 6–7) Let $A \in \mathbb{R}^{m\times n}$ have full row rank and assume there exists $x > 0$ with $Ax = b$. Consider

$$\min_{x>0} \sum_{i=1}^{n} x_i(\log x_i - 1) \quad \text{subject to } Ax = b.$$

Derive the Lagrange dual, prove uniqueness of the primal solution, and derive a Newton method in the dual variables. Prove that the Newton matrix is nonsingular and state the dominant cost per dense iteration.